

# ECO102: Introduction to Macroeconomics

Lecture 5: Monitoring Jobs and Inflation Contd.  
&  
Economic Growth (Chapter 23)

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# Outline

## ***Chapter 22: Monitoring Jobs and Inflation***

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# Alternative Price Indices

Due to the inherent bias that is embedded within CPI, other measures are often used as alternative price level index numbers:

- **Chained CPI:** A price index that is calculated using a similar method to that used to calculate chained-dollar real GDP.

It overcomes the sources of biasness in the CPI. It incorporates substitutions and new goods bias by using current and previous period quantities rather than fixed quantities from an earlier period

- **Personal Consumption Expenditure (PCE) Deflator:** The PCE Deflator is similar to the CPI but the basket of goods and services included in it is broader than that in the CPI as it includes all consumption expenditure, not only the items bought by a typical urban family

$$PCE\ Deflator = \frac{Nominal\ Consumption}{Real\ Consumption} \times 100$$

# Alternative Price Indices: GDP Deflator

- ***GDP Deflator:*** The GDP Deflator is similar to the PCE Deflator, however, it includes all the goods and services that are counted as part of GDP

It is an index of the prices of the items in consumption, investment, government expenditure, and net exports

$$GDP\ Deflator = \frac{Nominal\ GDP}{Real\ GDP} \times 100$$

***NOTE: By using the GDP deflator, we can deflate other nominal variables to find their real values. For example, the real wage rate is the nominal wage rate divided by the GDP deflator***

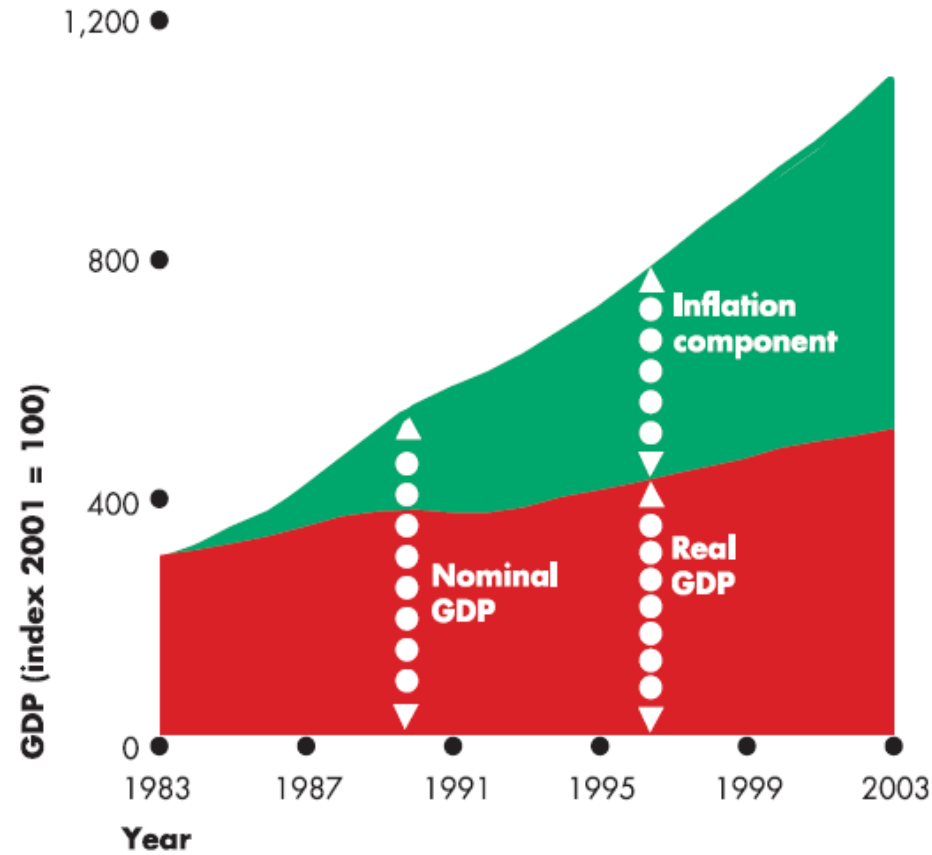
# Example

Calculate the GDP Deflator for the year ended in 2024 from the following information:

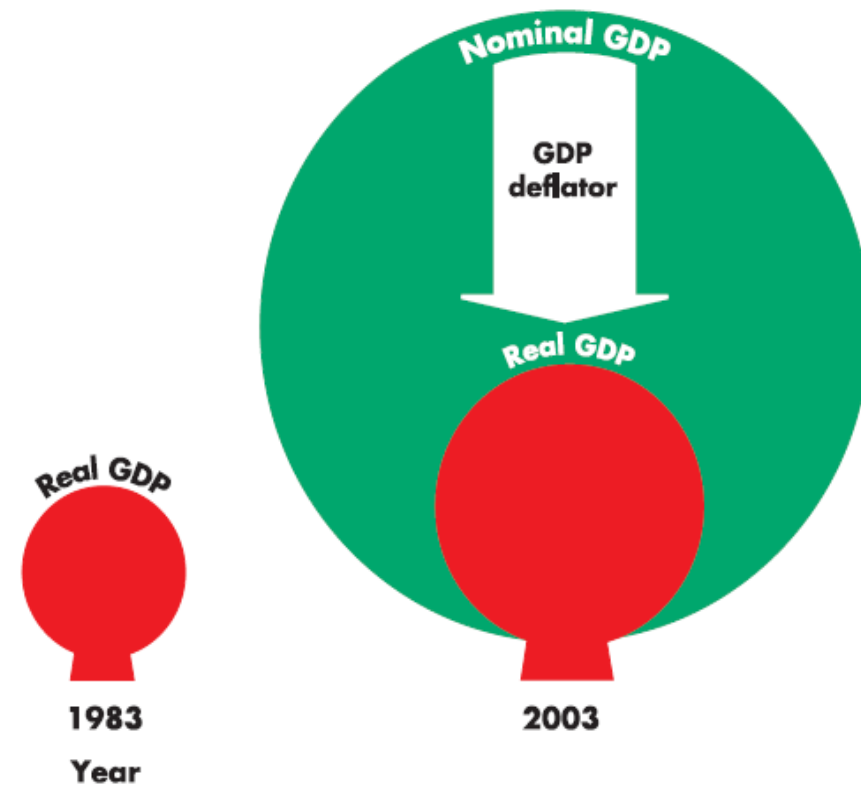
| Item                                        | Quantity<br>(2020) | Unit Price<br>(2020; US\$) | Quantity<br>(2024) | Unit Price<br>(2024; US\$) |
|---------------------------------------------|--------------------|----------------------------|--------------------|----------------------------|
| <b>T-Shirts</b>                             | 10                 | 5.00                       | 4                  | 5.00                       |
| <b>Private Investment in Computer Chips</b> | 3                  | 10.00                      | 2                  | 20.00                      |
| <b>Government Expenditure Public Health</b> | 1                  | 20.00                      | 6                  | 40.00                      |
| <b>Unemployment Benefits</b>                | 5                  | 15.00                      | 10                 | 20.00                      |

# Alternative Price Indices: GDP Deflator

The UK GDP Balloon



(a) Nominal GDP and real GDP



(b) Deflating nominal GDP

# Some Key Review Questions

- What is the difference between the working-age-population and the labour force?
- How do you define someone as not in the labour force?
- Who are marginally attached workers and who are discouraged workers? Who are economic part-time workers?
- Which is the most costly unemployment?
- How does the full employment connect to potential GDP
- What are some methods of calculating inflation?
- What is the primary difference between PCE Deflator and GDP Deflator?

# Chapter 23: Economic Growth



# Calculating Growth Rates

All of our discussions so far has been to understand the economic growth of different countries

Economic growth is a sustained expansion of production possibilities measured as the increase in real GDP over a given period

The ***economic growth rate*** of a country is expressed as the annual percentage change of ***real GDP***

$$\text{Real GDP Growth Rate} = \frac{\text{Real GDP in Current Year} - \text{Real GDP in previous year}}{\text{Real GDP in previous year}} \times 100$$

This formula will also be applicable when we want to calculate ***per capita Real GDP growth rate*** (for standard of living). We just replace the Real GDP values in the formula with per capita values.

# Review Problem

Assume for a given country “A”, the real GDP is \$11 trillion, and the population is 202 million in the current year. In the previous year, the real GDP for this country was \$10 trillion, and the population was 200 million. What is the Real GDP Growth Rate and the per capita Real GDP growth rate for this country?

# Sustained Growth

- A rapid and sustained economic growth has transformed the East Asian Countries in some of the major players of the global economy
- Countries like Japan, South Korea, Taiwan, Hong Kong, Singapore, Malaysia, Thailand, and Indonesia underwent rapid economic growth between the 1960s and 1990s. Often referred to as the East Asian Miracle, and these economies are termed Tiger Economies
- On the other hand slow economic growth or the absence of growth can condemn a nation to devastating poverty
- Nations like Sierra Leone, Somalia, and Zambia are some of the examples
- ***Sustained growth of real GDP per person can transform a poor society into a wealthy one. The reason is that economic growth is like compound interest***

# Question

*IF TECHNOLOGICAL PROGRESS DRIVES LONG-TERM GROWTH, WHY DO SOME COUNTRIES STILL LAG BEHIND DESPITE HAVING ACCESS TO SIMILAR TECHNOLOGIES?*